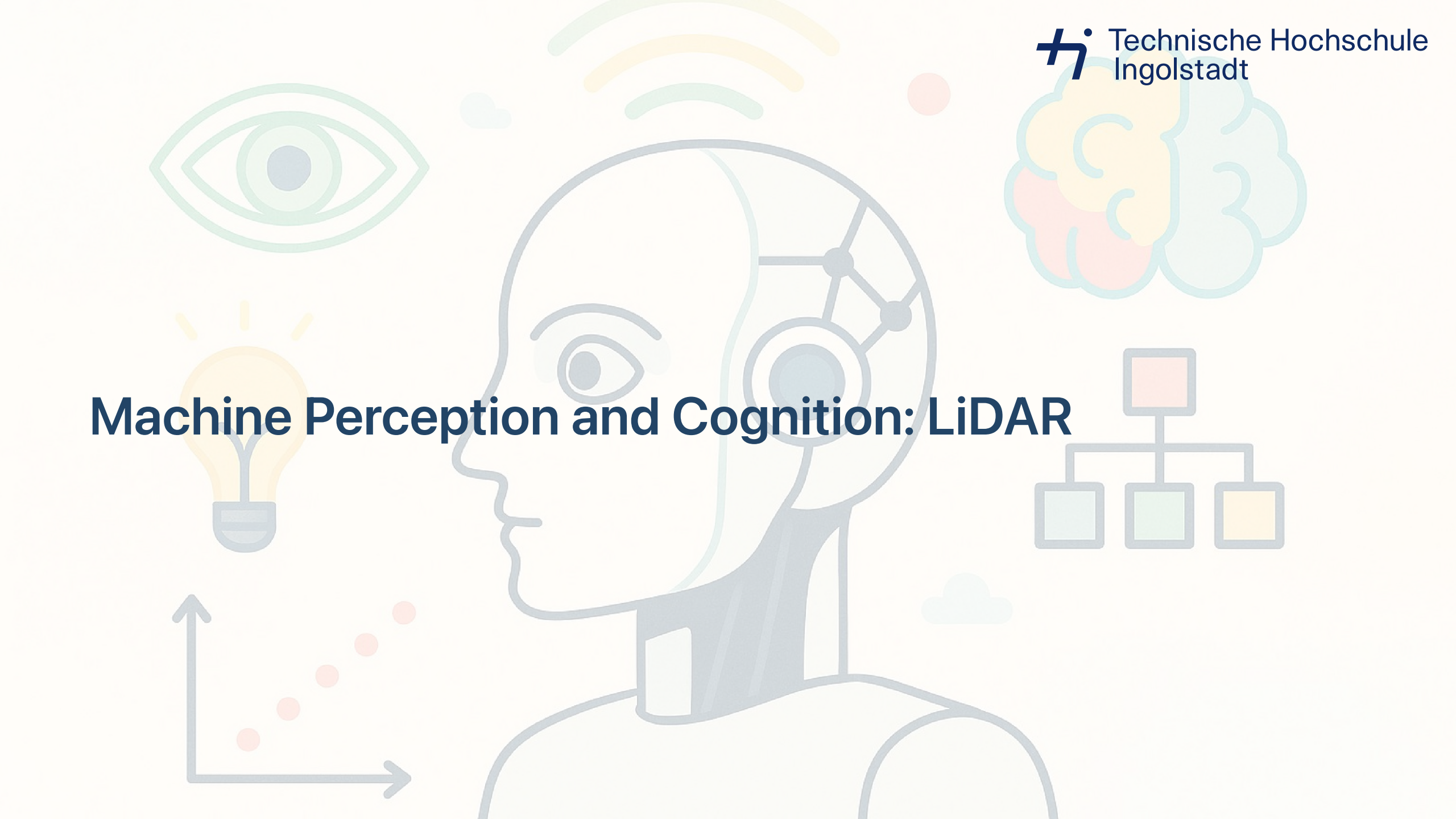
Technische Hochschule Ingolstadt



Machine Perception and Cognition: LiDAR



There seem to be two acronyms for LiDAR:

"Light detection and ranging" or "Laser imaging, detection and ranging".

Image source: https://www.pexels.com/photo/wooden-hand-holding-question-mark-10981245/

Why LiDAR if we have ToF?

Time-of-flight has the advantage in that it estimates the depth to the object per pixel. That is, it generates a RGBd depth map in a single snapshot.

Sounds pretty good, so what is the drawback?



Image source: https://www.pexels.com/photo/crop-person-using-smartphone-in-dark-room-3348724/



Why LiDAR if we have ToF?

How does a time-of-flight sensor capture depth information?

Image source: https://www.pexels.com/photo/long-exposure-photography-of-red-and-green-led-lights-3358953/

ToF limitations

A time-of-flight sensor is an active sensor that sends out a light pulse and measures the response (reflection). The light pulse covers the whole scene at once and thus is wide-angle and not focused.

Otherwise said: ToF uses active sensing with a wide-angle, low-intensity light pulse.

ToF limitations

The return signal intensity decreases quadratically with distance (inverse square law):

$$i = \frac{1}{d^2}$$

for intensity i and distance d.

And we have twice the distance that the light pulse has to travel!

ToF limitations

Other limitations of time-of-flight:

- 🎯 Limited spatial resolution, defined by the pixel grid.
- 📡 Susceptible for scattered signals and noise.
- 🔁 Multipath inference.
- ☀️ Sunlight is often stronger than ToF.

Onward to LiDAR

We need a high-intensity, focused light pulse for active sensing!



Image source: https://www.pexels.com/photo/the-cover-of-the-book-the-road-to-the-mountains-27769158/

Collimator

A collimator is a physical device to produce a narrow beam, here in a particle collimator.

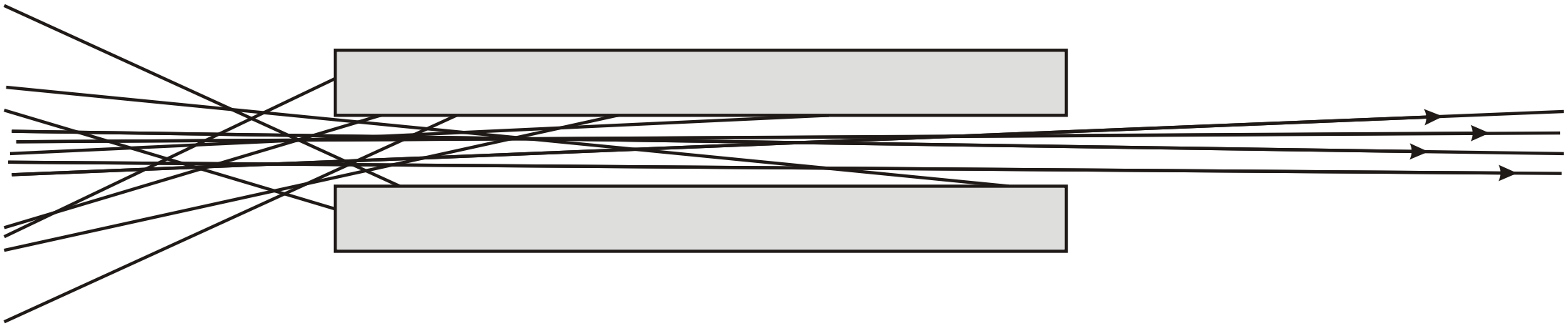
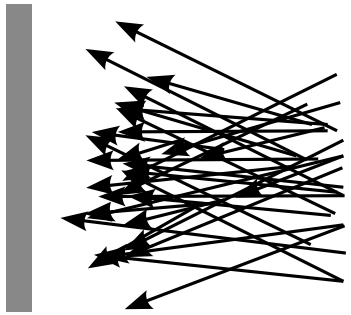


Image source: Mpfiz, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:ParticleCollimator.svg

Collimator

In a perfect collimator, paths of the beams should be perfectly parallel:



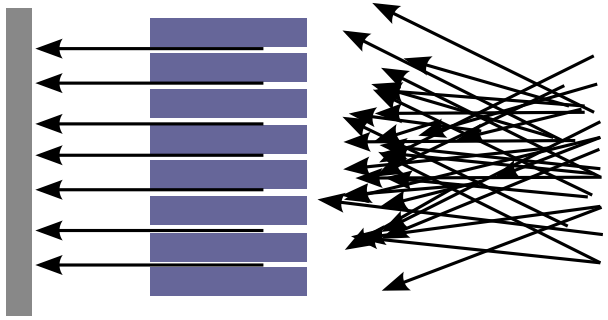


Image source: Original: Pete Verdon; Vector: Chris828, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Collimator2.svg

Collimator

A collimator can also shape the beam to create a wanted projection area:



Image source: Ikiwaner, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:UW_Collimator.jpg

Collimator

And you can build very large collimator devices:

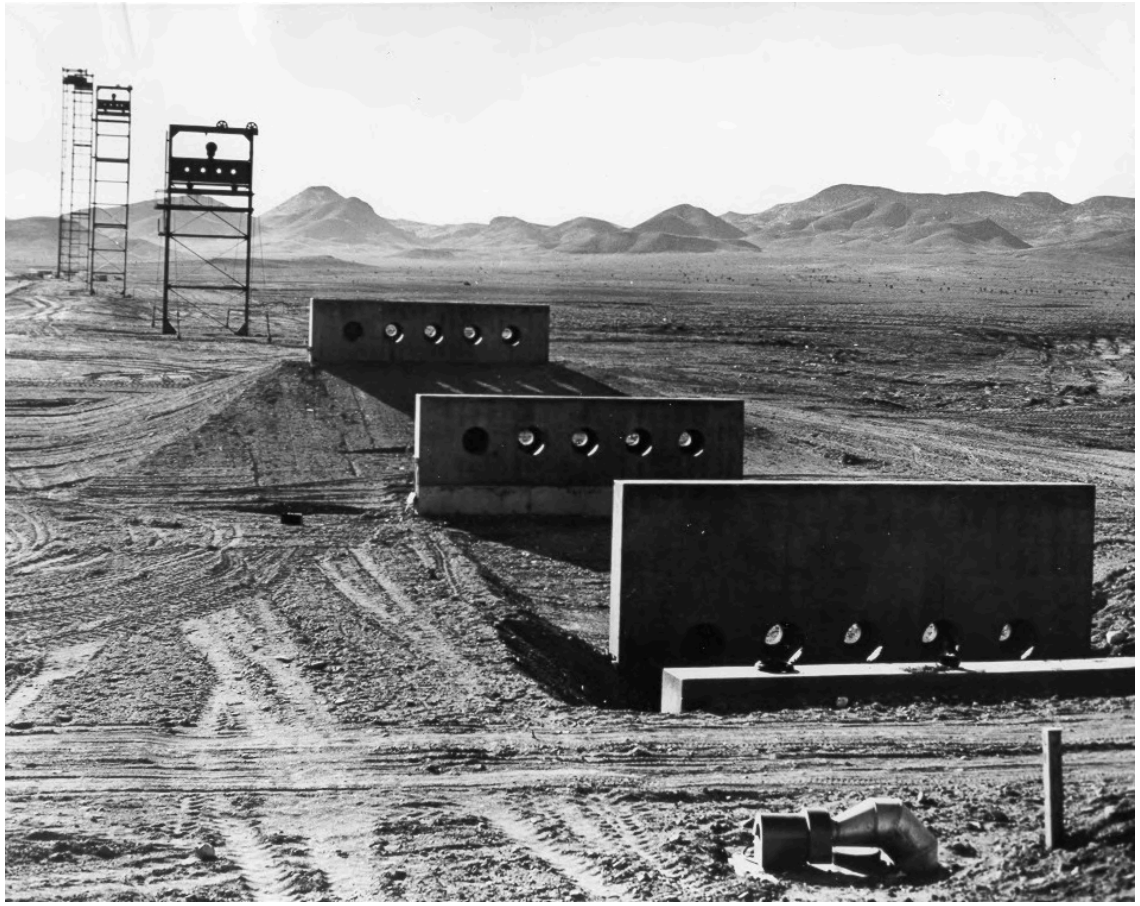


Image source: National Nuclear Security Administration / Nevada Site Office, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:NNSA-NSO-190.jpg

Collimator

Collimated light can be generated using optical lenses:

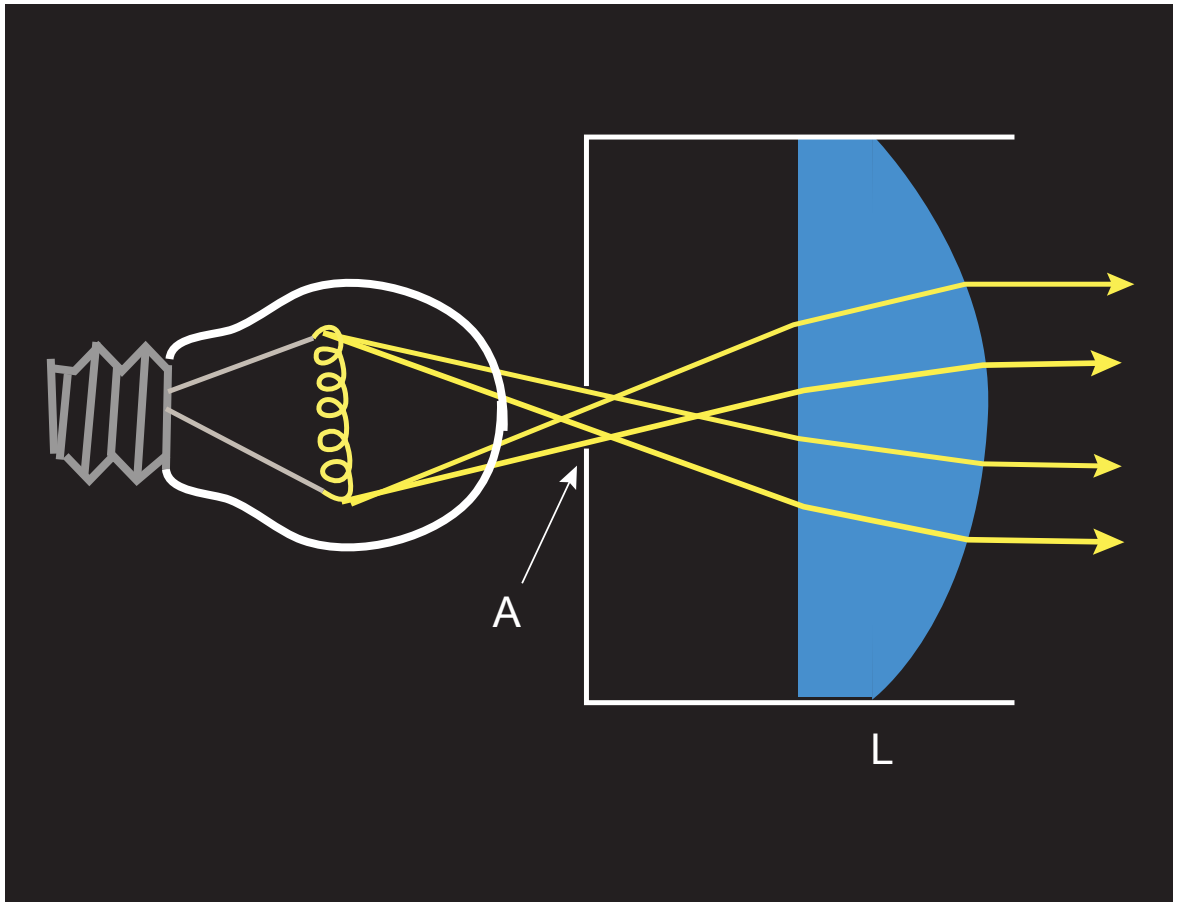


Image source: Krishnavedalal, CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Collimator.svg

Laser

We can also use a laser as a light source for our light pulse:



Image source: ESO/A. Ghizzi Panizza (www.albertoghizzipanizza.com), CC BY 4.0 (https://creativecommons.org/licenses/by/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Guiding_the_Milky_Way_(potw2222a).jpg

Laser

Lasers produce light with low divergence in directionality and nearly a single wavelength:

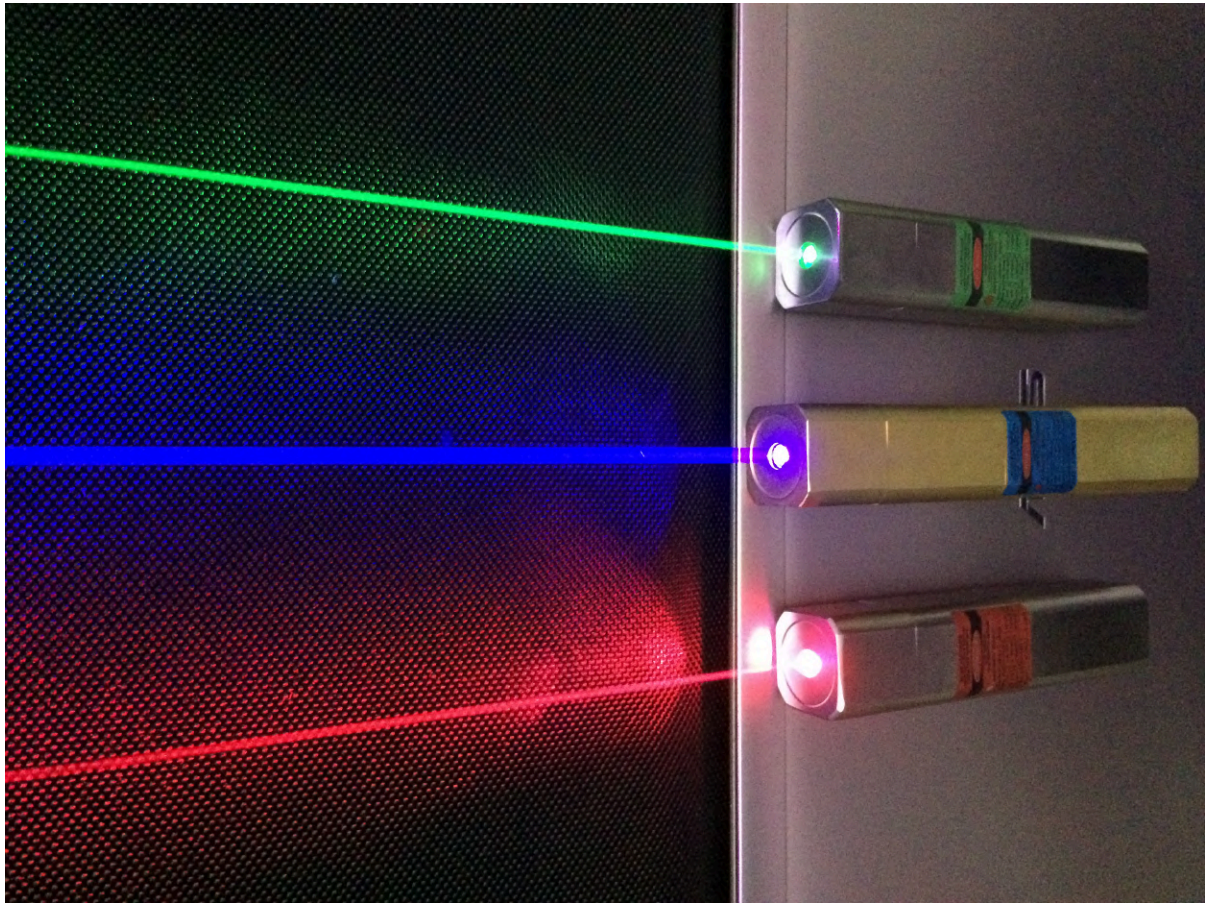


Image source: , CC BY-SA 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Lasers.jpeg

LiDAR improvements

LiDAR uses collimated light from a laser as source:

- 💡 Higher initial intensity possible, think inverse square law.
- 🔦 Focused light with less scattering.
- 🎯 Nearly a single wavelength allows to filter out other light sources.

Drawback: LiDAR scans points sequentially, building a point cloud over time.

How does LiDAR work?

Basic principle as in time-of-flight cameras:

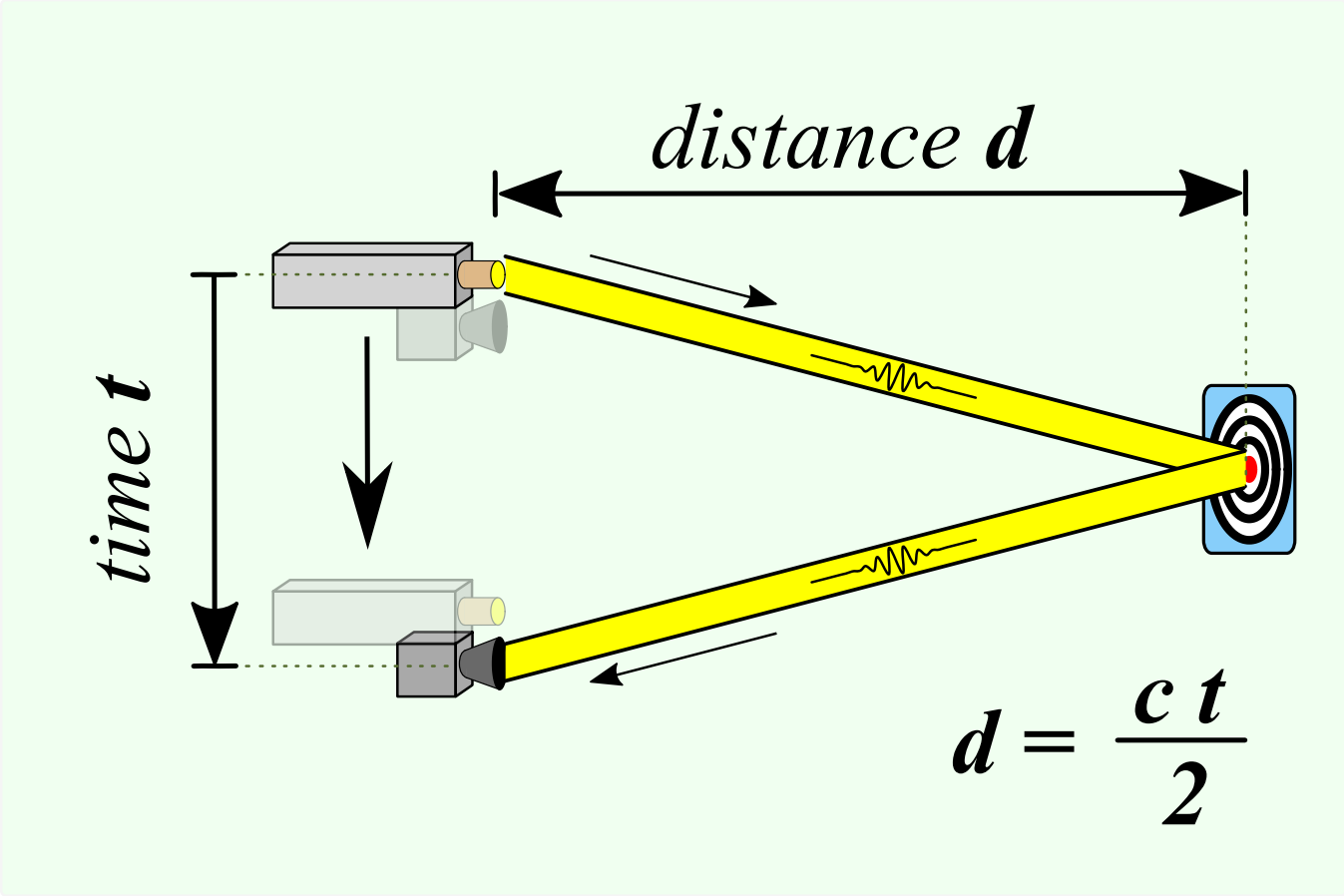


Image source: RCraig09, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:20200501_Time_of_flight.svg

How does a LiDAR work?

Animation of how a LiDAR scans its surroundings:

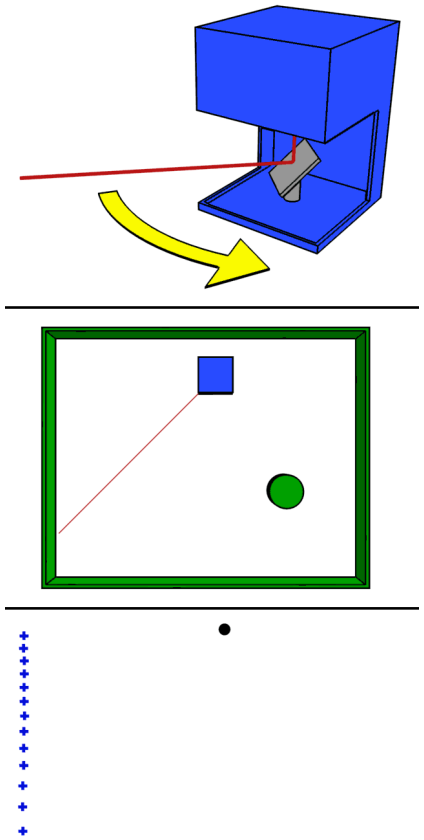


Image source: Mike1024, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:LIDAR-scanned-SICK-LMS-animation.gif

How does a LiDAR work?

In a LiDAR, we expect a narrow/focused return signal that is still very weak in intensity because of the inverse square law. Since we are expecting a return signal only at a single point, not a matrix, we can use some specialized hardware:

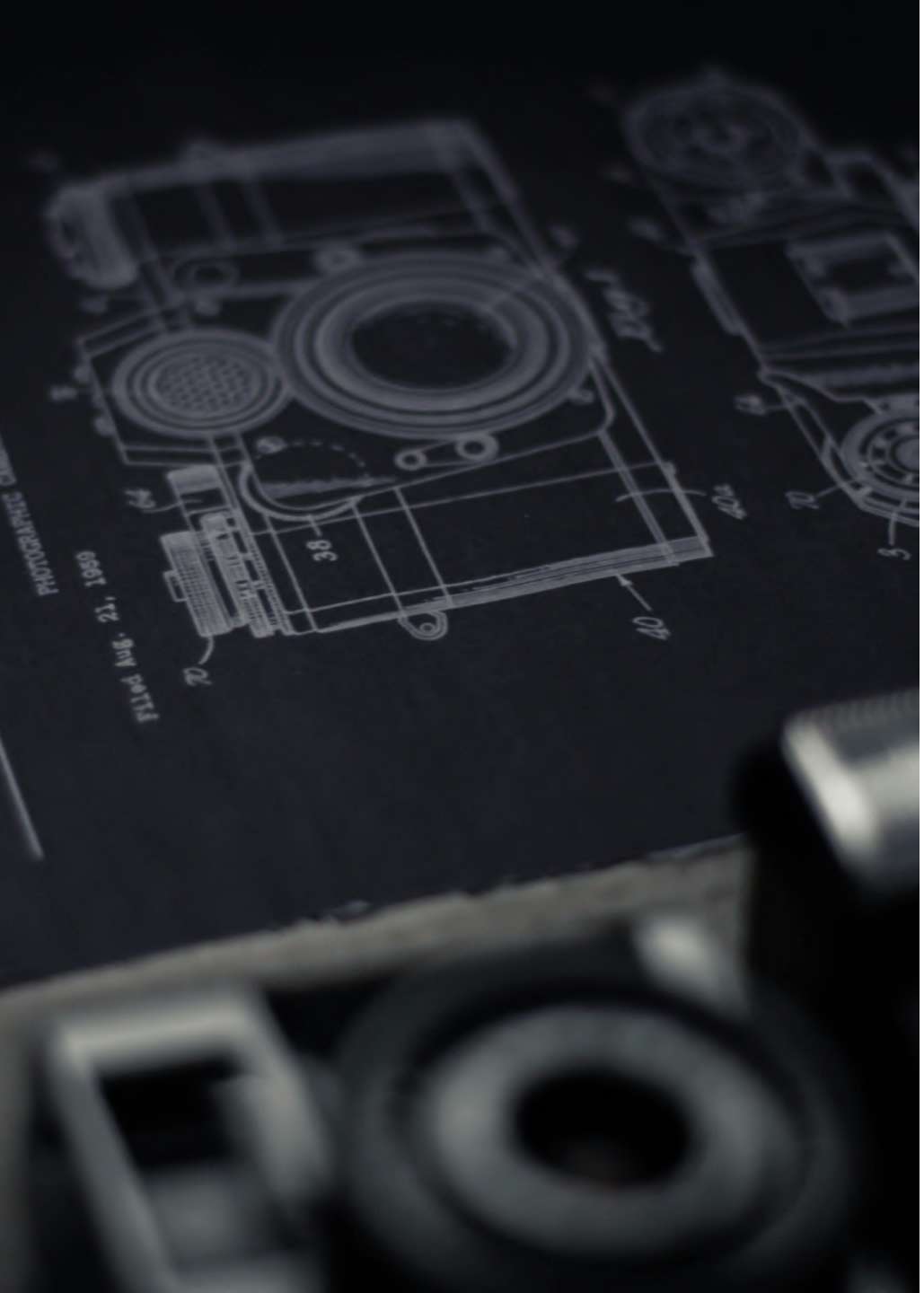
LiDAR often uses an *avalanche photodiode* or a *single-photon avalanche diode*. There can also be multiple diodes organized in a raster like in a CCD or CMOS.

Photon counter!

Photon counters allow to handle photons as events: photon = 1, no photon = 0.



Image source: Geek3, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Single-Photon-Counting-Module_Excelitas.jpg



Photon counter!

Did a Google Patents search and the patent for the above photon counter was filed in 2001...

Image source: https://www.pexels.com/photo/scheme-of-modern-photo-camera-4172287/

Time-to-digital converter

From Wikipedia, page "Time-to-digital converter":

> In electronic instrumentation and signal processing, a **time-to-digital converter (TDC)** or **time digitizer (TD)** is a device for recognizing events and providing a digital representation of the time they occurred. For example, a TDC might output the time of arrival for each incoming pulse. Some applications wish to measure the time interval between two events rather than some notion of an absolute time, and the digitizer is then used to measure a time interval and convert it into digital (binary) output. In some cases an interpolating TDC is also called a **time counter (TC)**.



Time-to-digital converter

Time-to-digital converter can have resolutions in picoseconds ($10^{-12}s$).

Image source: https://www.pexels.com/photo/close-up-photo-of-analog-clocks-5563414/

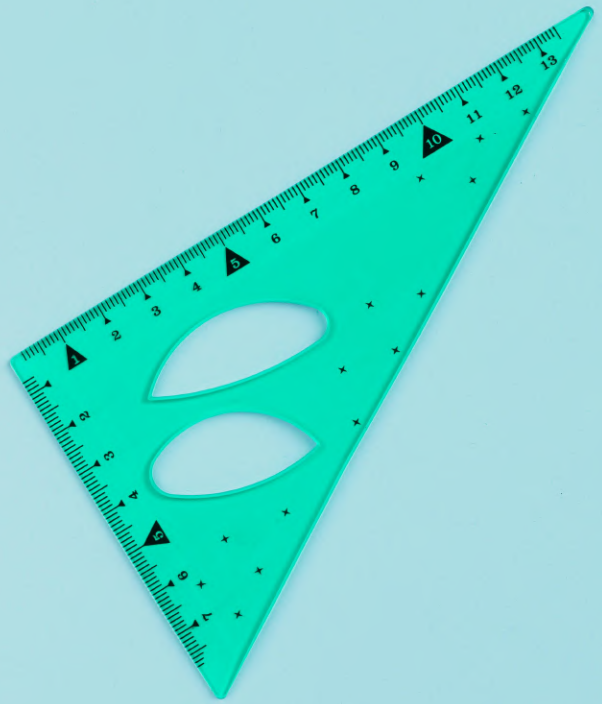
How does a LiDAR work?

So now we can actually put the LiDAR together:

1. Laser (with collimator) produces a high-intensity, narrow, monochromatic light pulse.
2. Light pulse is targeted towards a specific point which is scanned at that moment.
3. Light pulse is reflected off the surface of the scanned object.
4. Light pulse is received by the (single-photon) avalanche diode.
5. Time between emitting and receiving the light pulse is measured with high precision.
6. Calculate the distance and repeat with the next point to be scanned!

Realistic distanced for LiDAR

Type of LiDAR	Minimum distance	Maximum distance
Robotics	~10cm	~10-30m
Automotive, drones	~15cm	~100-200m
Topographic, airborne, space-based	~50cm	> 10km



Angular resolution

So far we have talked about spatial or temporal resolution in sensors. A LiDAR has an angular resolution.

Why? Surroundings are scanned point-by-point using a rotating mirror to direct the light pulse.

Image source: https://www.pexels.com/photo/set-sqaure-on-blue-background-5412366/

Angular resolution

Yang, D., Liu, Y., Chen, Q., Chen, M., Zhan, S., Cheung, N. K., ... & Li, W. J. (2023). Development of the high angular resolution 360 LiDAR based on scanning MEMS mirror. *Scientific Reports, 13*(1), 1540:

> The experimental results verified an excellent performance of 0.07° × 0.027° (horizontal × vertical) angular resolution, which enhances the panoramic scanning and imaging capability of the LiDAR system, potentially providing more accurate 3D scanning applications in areas such as autonomous vehicles, indoor surveying, indoor robotics navigation, etc.

Angular error

In addition, there is an angular error because of environmental conditions such as atmospheric scattering, haze, smoke or sunlight.



Image source: https://www.pexels.com/photo/fluffy-white-clouds-against-a-bright-blue-sky-29008111/

Angular error

So how much error does a LiDAR produce? We need: $\tan\theta = \frac{opposite}{adjacent}$

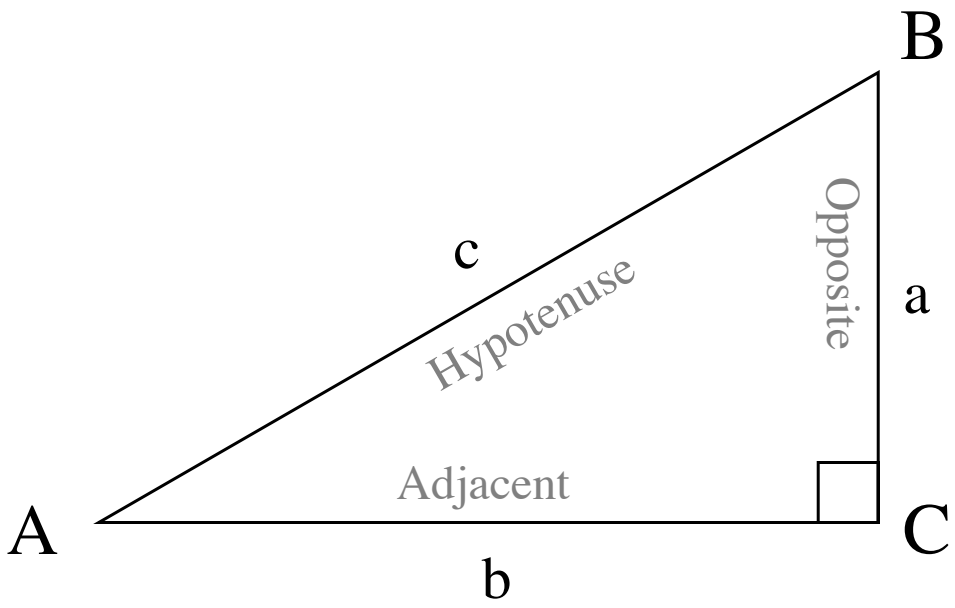


Image source: TheOtherJesse, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:TrigonometryTriangle.svg

Angular error

Let $\theta = 0.07^\circ$ or $\theta = 0.07^\circ \times \frac{\pi}{180^\circ} \approx 0.001222$ be angular error and $d = 100m$ be distance, then error e is:

$$e = d \times \tan\theta \approx 100m \times \tan 0.001222 \approx 100m \times 0.001222 \approx 0.12m$$

LiDAR devices

LiDAR in robotics:



Image source: S. Winkvist, Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:LIDAR_equipped_mobile_robot.jpg

LiDAR devices

LiDAR in a UAV:



Image source: Jonte, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Drone_with_Lidar.jpg

LiDAR devices

LiDAR in an autonomous vehicle:



Image source: 9yz, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Zoox_Toyota_Highlander_Test_Vehicle_-_San_Francisco,_May_2025_08.jpg

LiDAR devices

A LiDAR is much a mechanical device:



Image source: Bunyk, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:LDS02RR_LIDAR_bearing_and_motor.jpg

LiDAR examples

Height map of the surface of the moon from the Clementine mission:

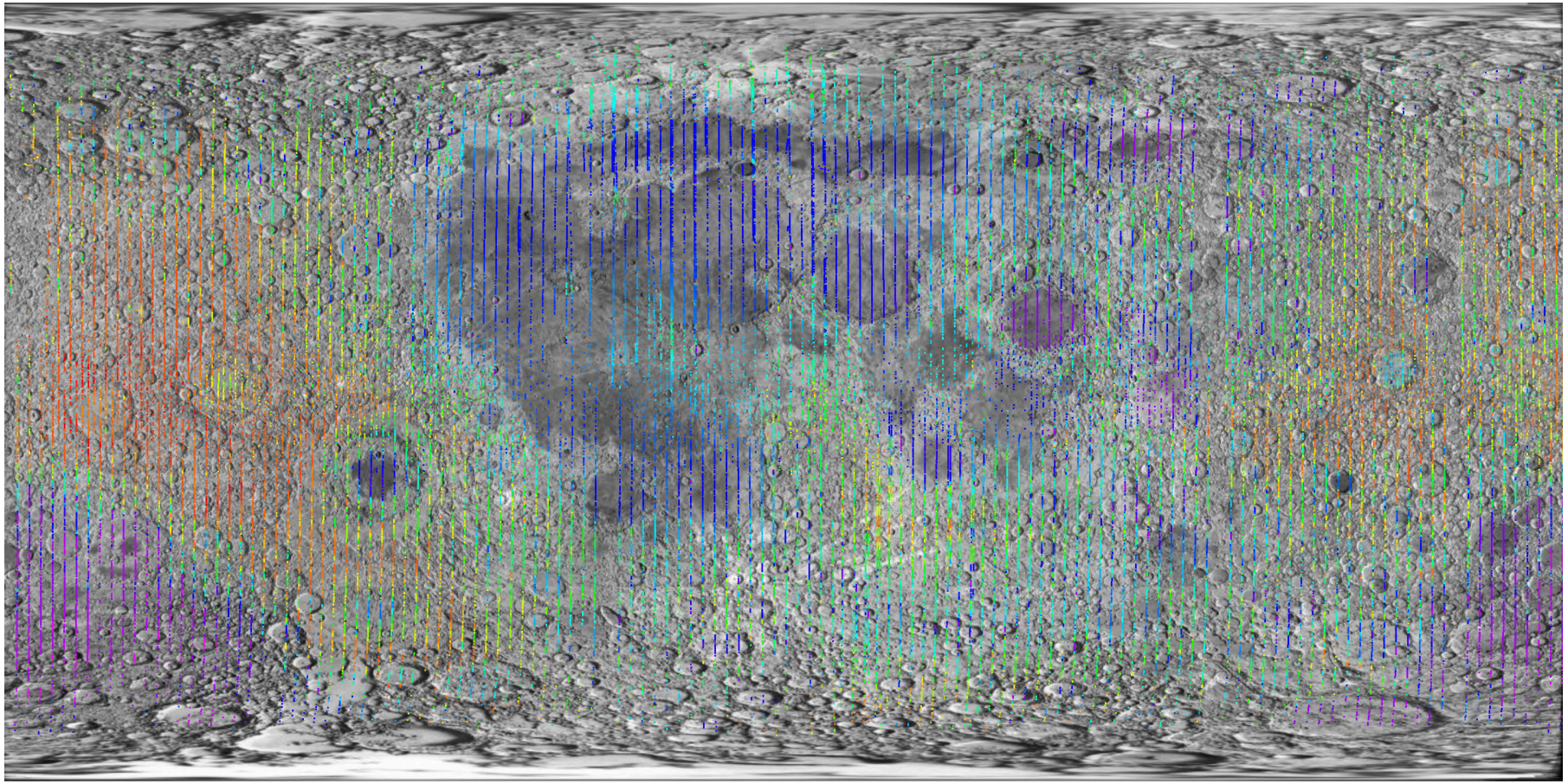


Image source: Martin Pauer (Power), Public domain, via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Moon_clementine_lidar.jpg

LiDAR examples

Highway intersection:

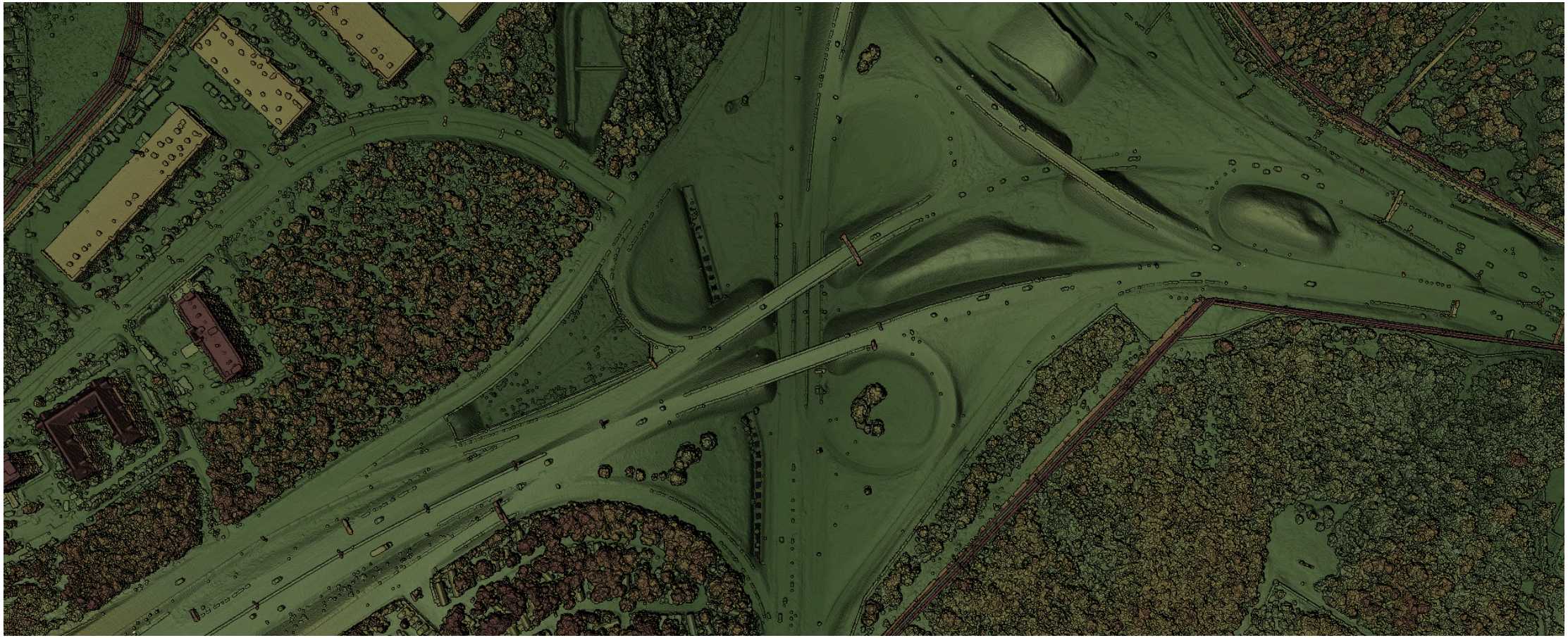


Image source: Rebecca equator, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Austin_Texas_LiDAR_Point_Cloud_(Highway_183).png

LiDAR examples

Street intersection:

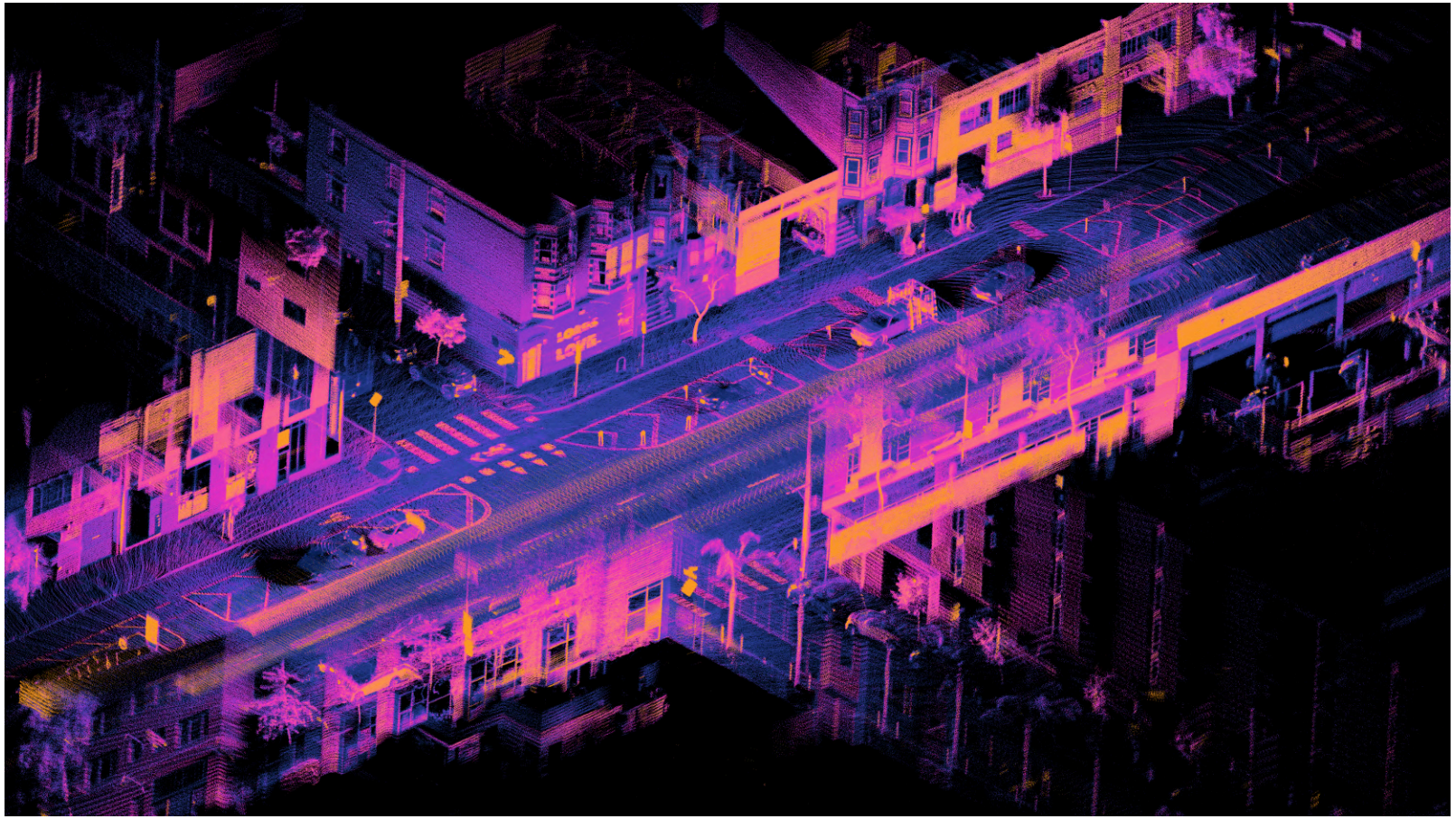


Image source: Daniel L. Lu, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Ouster_OS1-64_lidar_point_cloud_of_intersection_of_Folsom_and_Dore_St,_San_Francisco.png

LiDAR examples

Mechanical room:

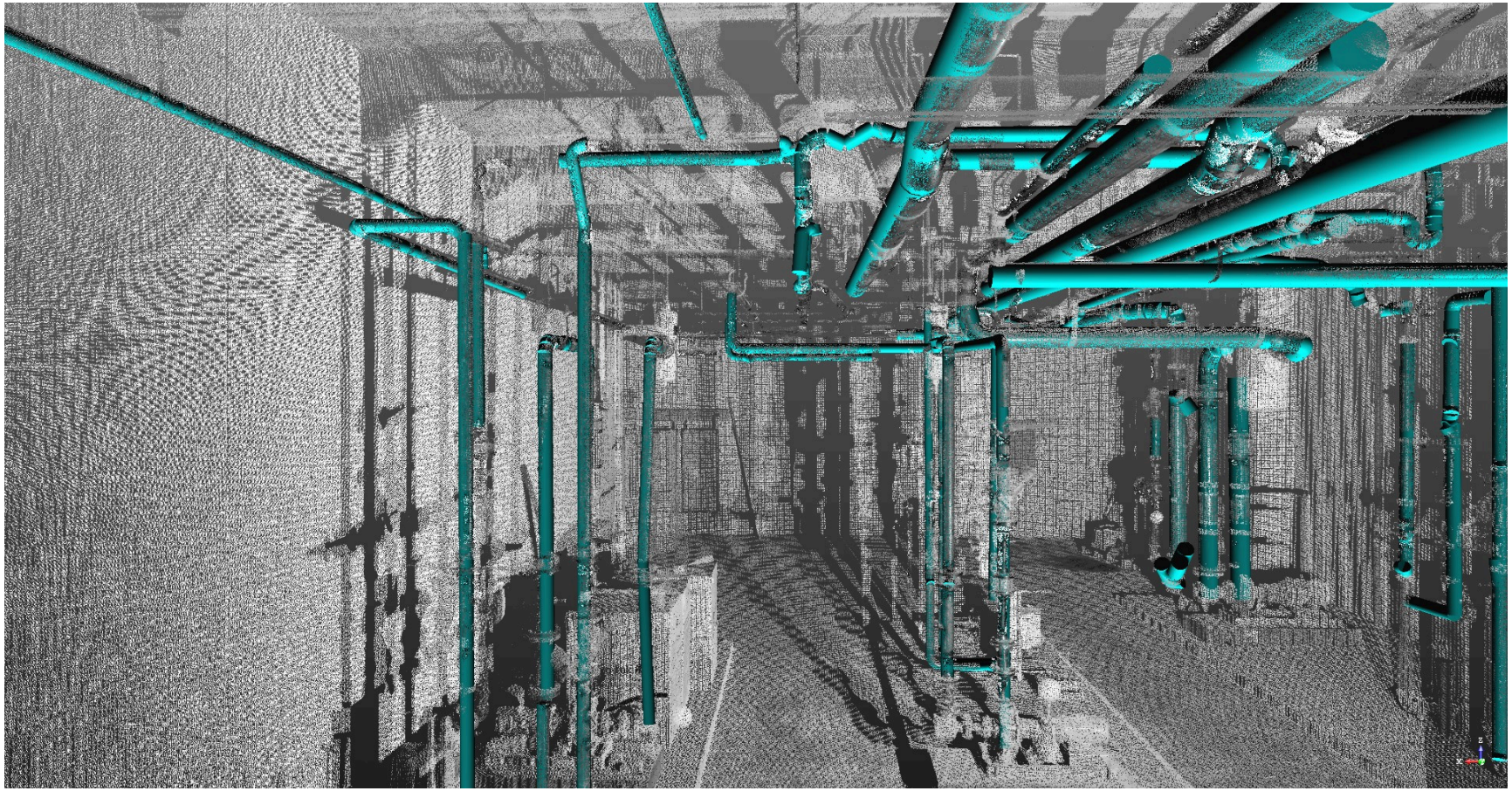


Image source: Oregon State University, CC BY-SA 2.0 (https://creativecommons.org/licenses/by/2.5), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Scan-to-BIM_Mechanical_Room.jpg

LiDAR examples

Airport passport control:

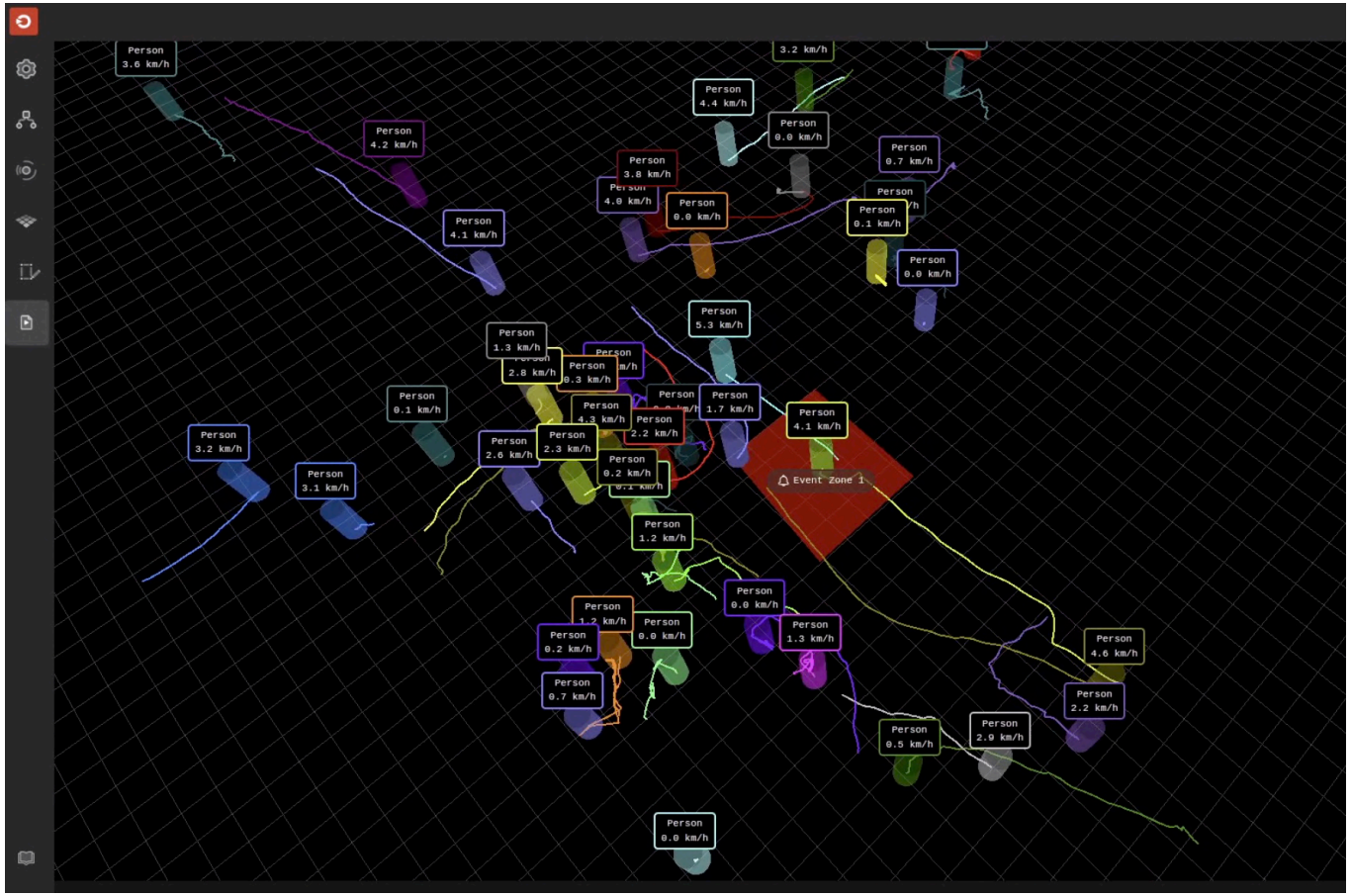


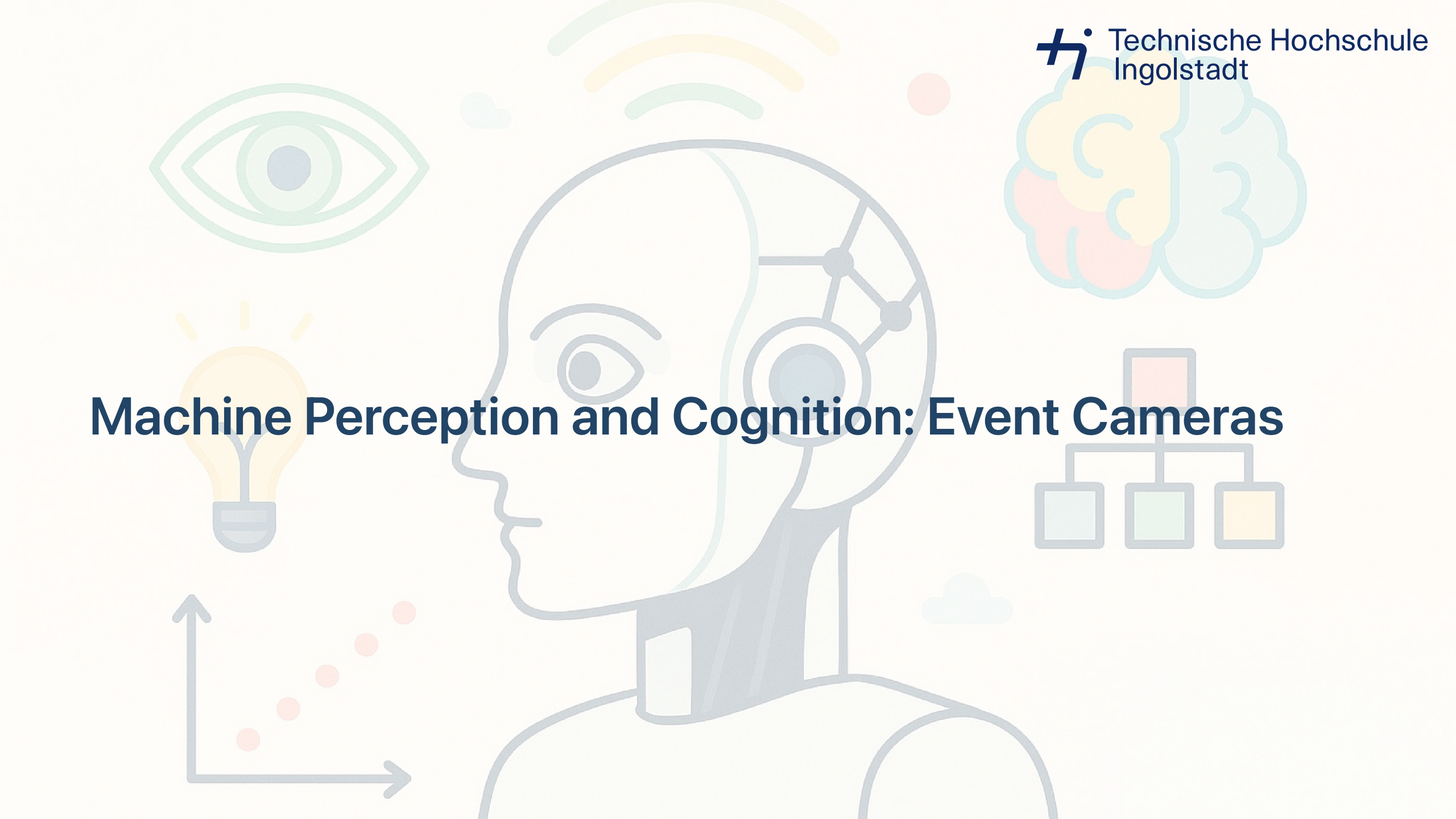
Image source: Saxotango, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:People_Tracking_with_LiDAR_in_Airport_Passport_Control.jpg

LiDAR

Recap!

- 🔦 Active depth sensing using a collimated laser and time-of-flight measurement.
- 🎯 Produces high-range, precise point measurements to build 2.5D depth map.
- 📉 Advantages over ToF: Longer range and less noise.
- 🔧 Key components: Laser (with collimator), (single-photon) avalanche photodiode, time-to-digital converter, scanning mirror.
- ⚠️ Limitations: Angular resolution, atmospheric scattering, inverse-square signal loss.

Technische Hochschule Ingolstadt



Machine Perception and Cognition: Event Cameras



What's an event camera?

Image source: https://www.pexels.com/photo/question-mark-on-crumpled-paper-5428826/

What's an event camera?

CCD or CMOS sensors produce images of light intensities, organized in a rasterized image. Light intensities per pixel are measured either for the full frame at once or row-by-row. The image then is a rasterized representation of the absolute light intensity per pixel.

Event cameras report changes in light intensity, not absolute values! They do this for asynchronously for individual changes, not for the full frame at once.



Why do this?

CCD and CMOS sensors have a speed limit: How many frames can be captured per second?

Longer exposure times introduce motion blur. Row-wise shutters can lead to negative effects.

Image source: https://www.pexels.com/photo/road-sign-with-maximum-speed-limit-30-30084212/

Photosensitive capacitor

One way to build an event camera is with a photosensitive capacitor:

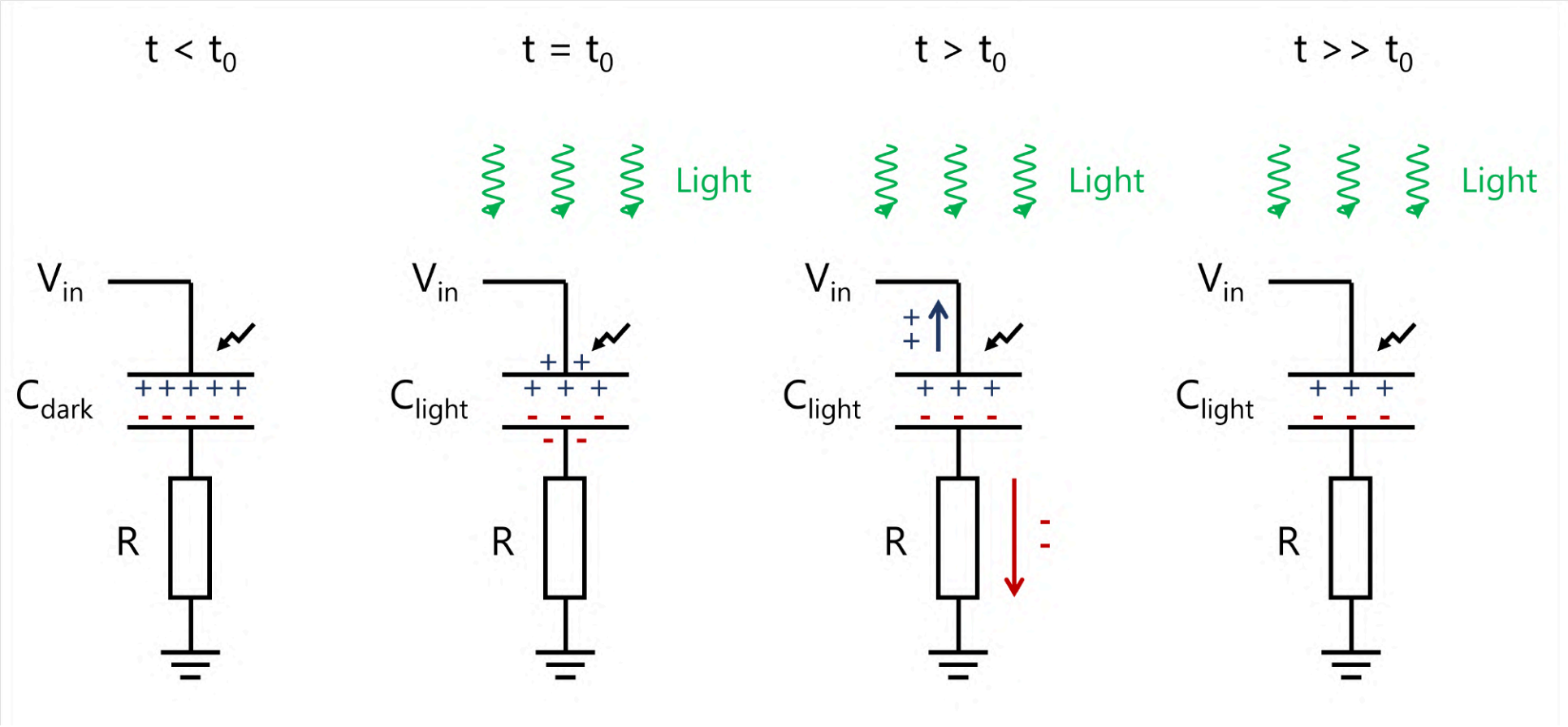


Image source: NoAppliedVoltage, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://en.wikipedia.org/wiki/File:Charge_on_Photosensitive_Capacitor.png



Photosensitive capacitor

A photosensitive capacitor changes its maximum electrical charge it can store in relation to higher or lower light intensities. Excess charge then needs to be discharged or additional charge can be accumulated.

Image source: https://www.pexels.com/photo/black-computer-compenent-159201/

Photosensitive capacitor

Light incoming on the photosensitive capacitor will change its capacity:

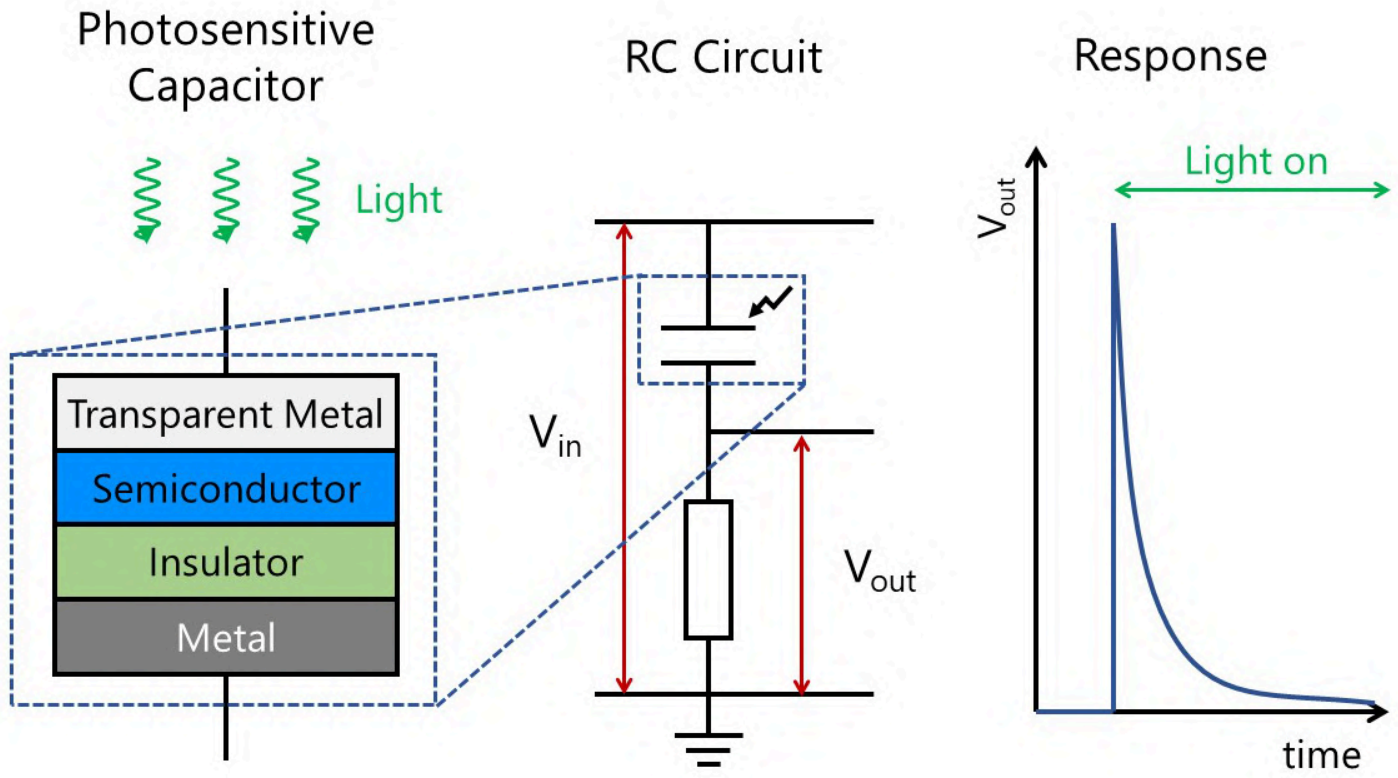
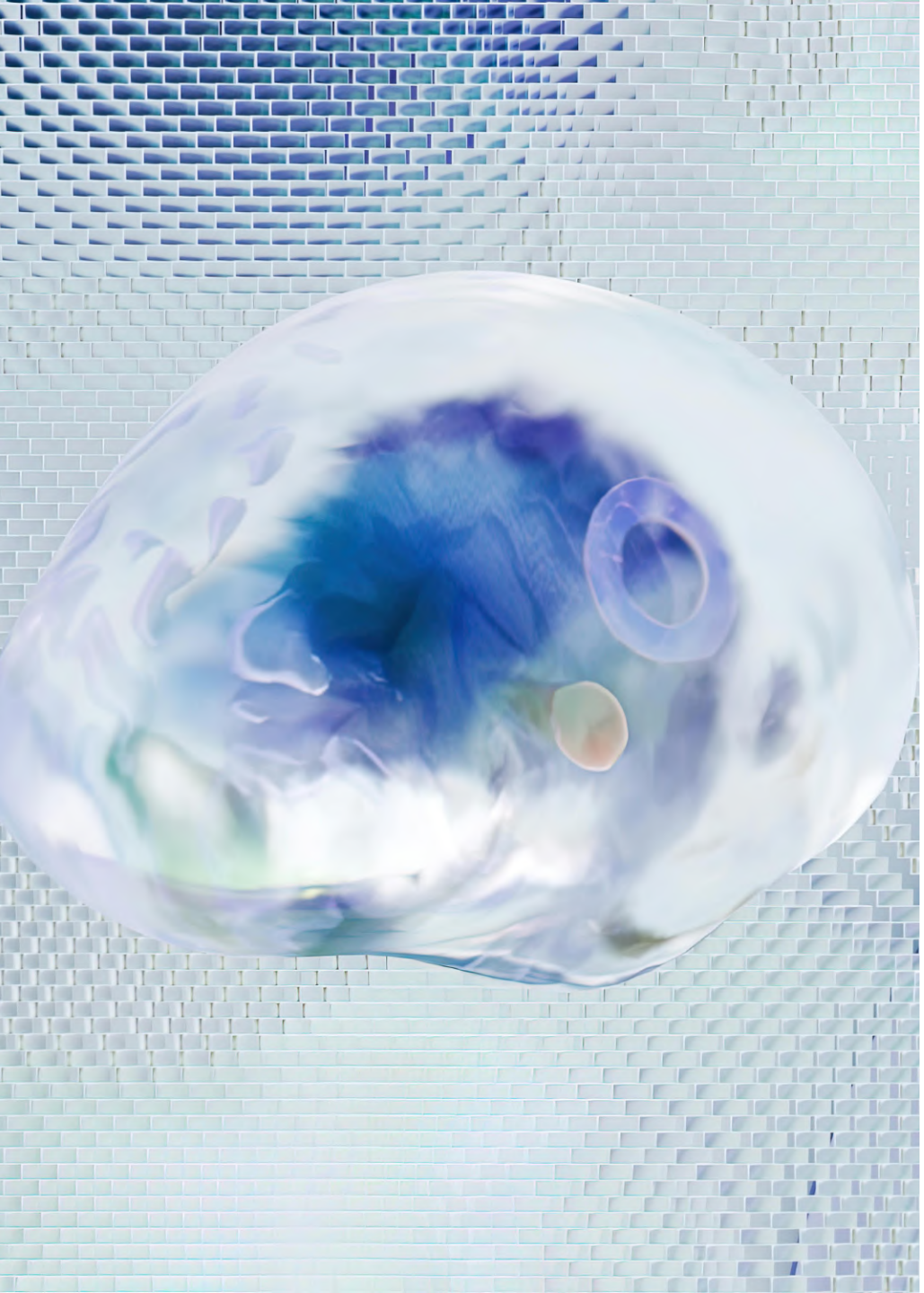


Image source: NoAppliedVoltage, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Retinomorphic_Sensor.jpg

Retinomorphic sensor

From Wikipedia, page "Retinomorphic sensor":

> **Retinomorphic sensors** are a type of event-driven optical sensor which produce a signal in response to changes in light intensity, rather than to light intensity itself. This is in contrast to conventional optical sensors such as charge coupled device (CCD) or complementary metal oxide semiconductor (CMOS) based sensors, which output a signal that increases with increasing light intensity. Because they respond to movement only, retinomorphic sensors are hoped to enable faster tracking of moving objects than conventional image sensors, and have potential applications in autonomous vehicles, robotics, and neuromorphic engineering.



Retinomorphic sensor

A retinomorphic sensor is one way to build an event camera. It is a pixel array like in a CCD or CMOS, but individual pixels react to changes in light intensity and do not measure absolute light intensity. These sensors are designed in such a way that they respond to changes in light intensities, creating an event with timestamp and pixel address.

Image source: https://www.pexels.com/photo/an-artist-s-illustration-of-artificial-intelligence-ai-this-image-represents-how-machine-learning-is-inspired-by-neuroscience-and-the-human-brain-it-was-created-by-novoto-studio-as-par-17483869/

Event camera

Comparison between an event camera and frame-wise camera:

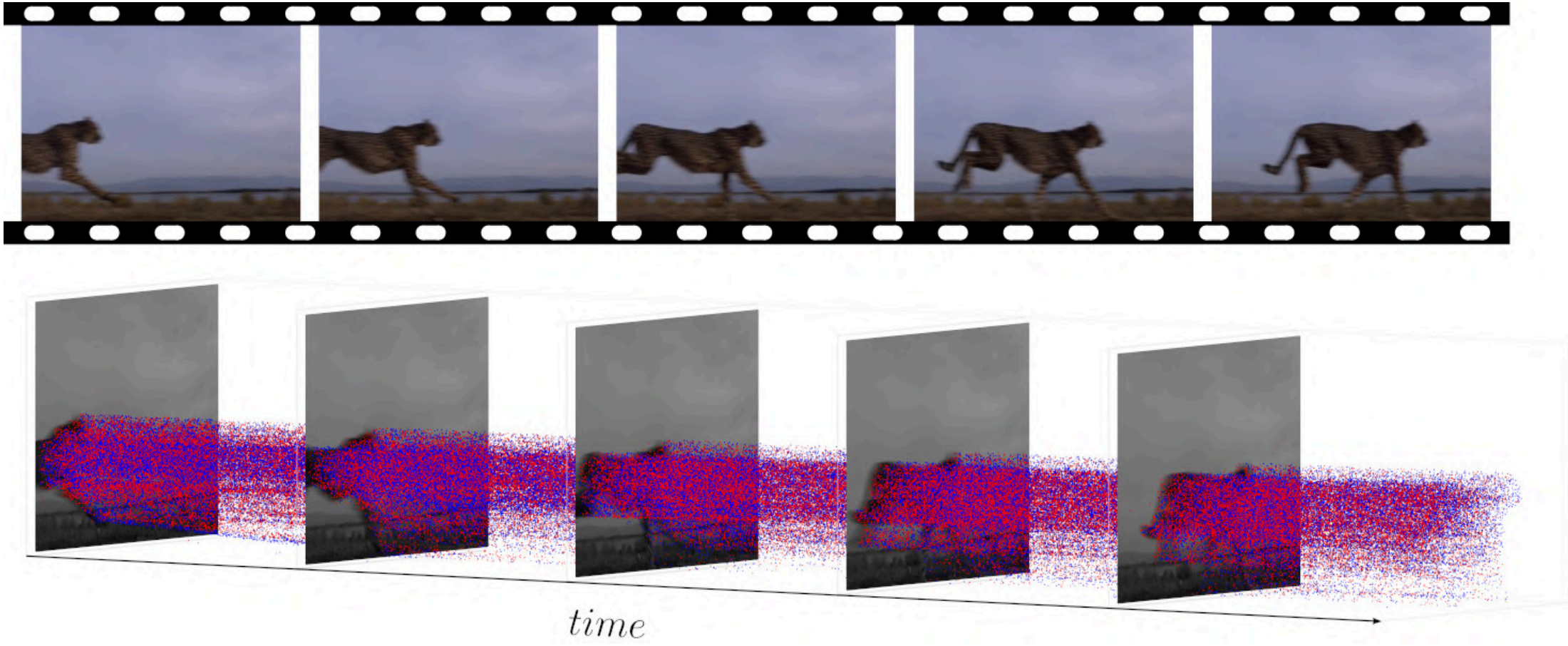


Image source: TimoStoffregen, CC BY-SA 4.0 (https://creativecommons.org/licenses/by-sa/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Event_camera_comparison.jpg

Event camera

Frame-wise images reconstructed from events by updating with the last received events:

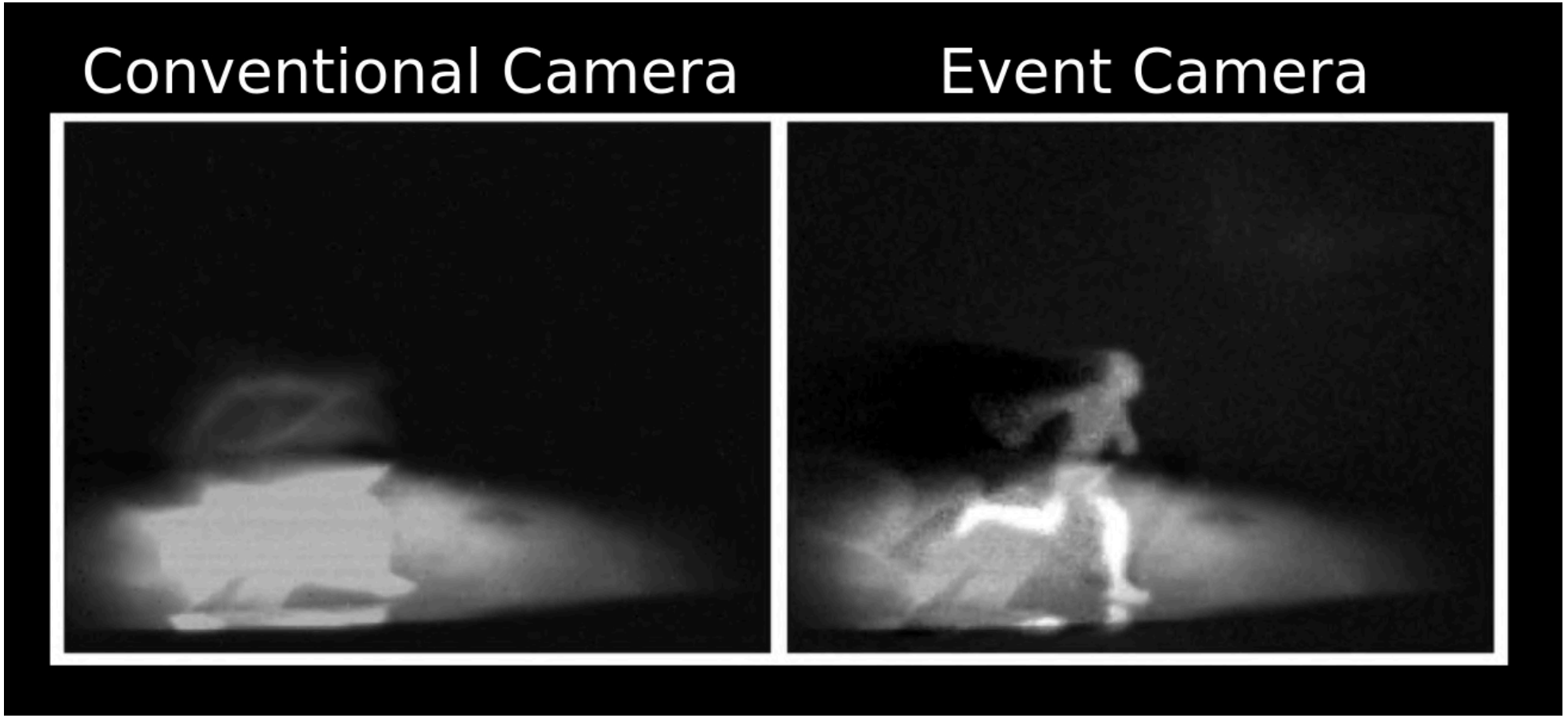


Image source: Cedric Scheerlinck, MIT , via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Night_run_reconstruction.png

Event camera

Performance of an event camera, from Wikipedia, page "Event camera":

Sensor	Dynamic range (dB)	Equivalent framerate (FPS)	Spatial resolution (MP)
Human eye	30-40	200-300	-
High-end DSLR camera (Nikon D850)	44.6	120	2-8
Ultrahigh-speed camera (Phantom v2640)	64	12500	0.3-4
Event camera	120	50000-300000	0.1-1

Limits of human visual perception

Tinsley, J. N., Molodtsov, M. I., Prevedel, R., Wartmann, D., Espigulé-Pons, J., Lauwers, M., & Vaziri, A. (2016). Direct detection of a single photon by humans. *Nature communications, 7*(1), 12172.:

> Despite investigations for over 70 years, the absolute limits of human vision have remained unclear. Rod cells respond to individual photons, yet whether a single-photon incident on the eye can be perceived by a human subject has remained a fundamental open question. Here we report that humans can detect a single-photon incident on the cornea with a probability significantly above chance.

Event camera device



Image source: Auledas, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Prophesee_Event_Camera_Evaluation_Kit.jpg

Why event cameras?

- 🚫 No motion blur.
- ⚡ Much faster reaction times.
- 🌓 High dynamic range.
- 🔁 Principle closely connected with spiking neural networks.

Spiking neural network

Neural networks as we use them (mostly) today encode value as amplitude.

Spiking neural networks encode value as higher frequencies of "spikes".

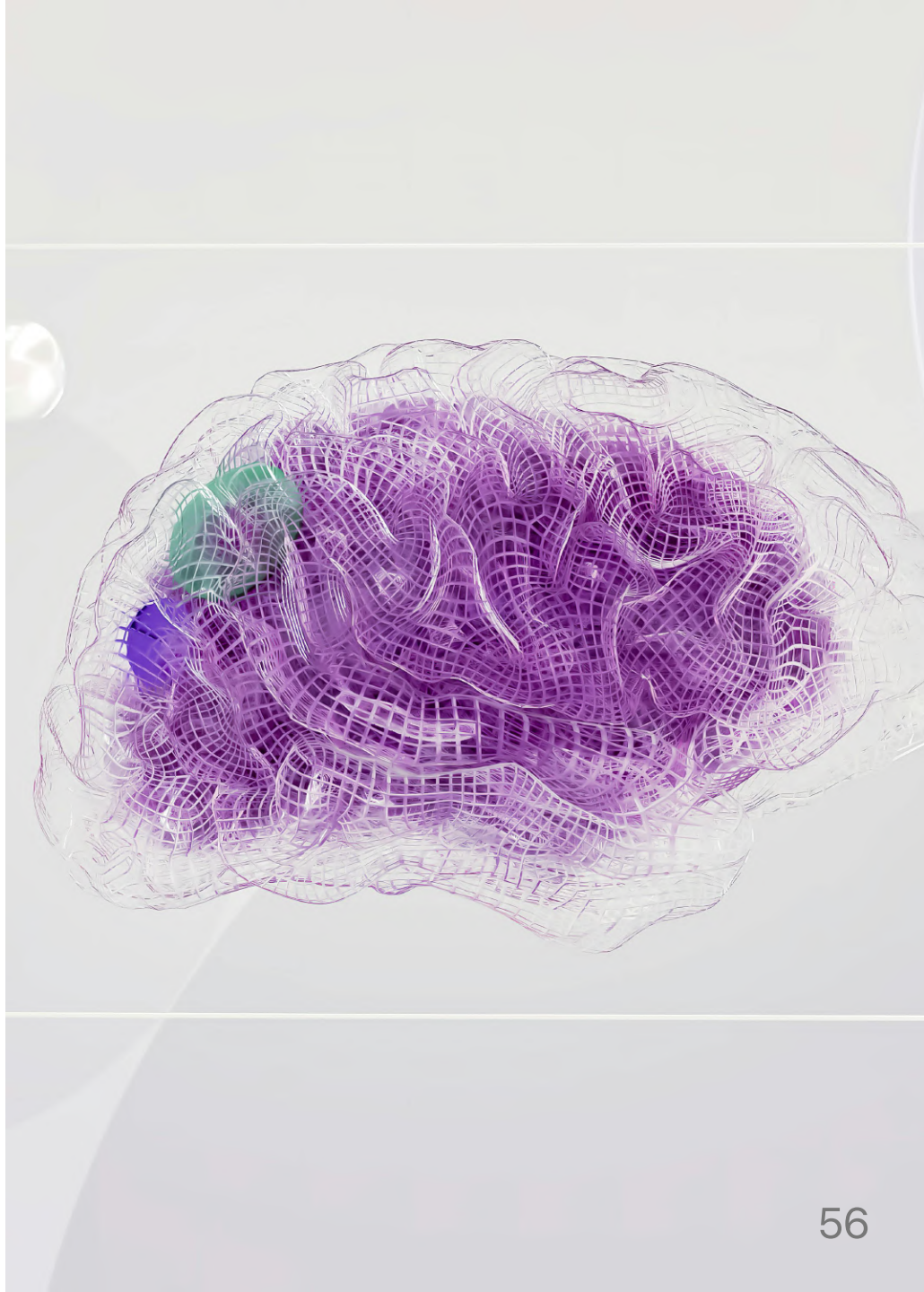


Image source: https://www.pexels.com/photo/an-artist-s-illustration-of-artificial-intelligence-ai-this-image-represents-how-machine-learning-is-inspired-by-neuroscience-and-the-human-brain-it-was-created-by-novoto-studio-as-par-17483868/

Spiking neural network

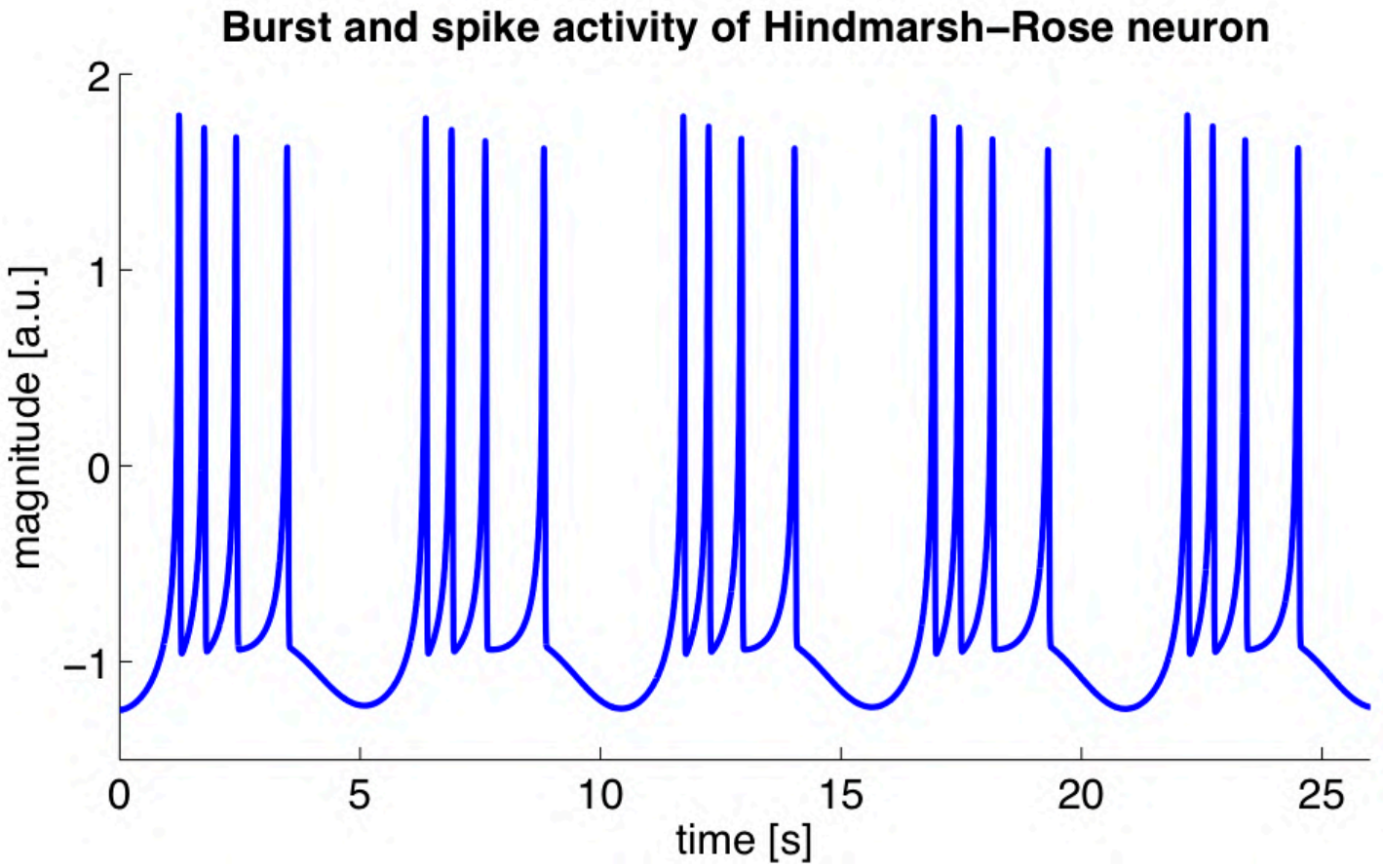


Image source: TjeerdB at English Wikipedia, CC BY 3.0 (https://creativecommons.org/licenses/by-sa/3.0), via Wikimedia Commons from https://commons.wikimedia.org/wiki/File:Simulation_of_hrose_neuron.png



Why event cameras?

Event camera seem to go "hand in hand" with spiking neural networks. Changes in light intensities produce spikes, which can be input into a spiking neural network.

Image source: https://www.pexels.com/photo/illuminated-light-bulb-2157840/

Event cameras

Recap!

- 🔦 Event cameras react to changes in light intensity, not absolute intensity.
- 🔁 Changes in light intensity are asynchronously reported as events.
- 📍 Events contain the pixel address and timestamp of the change.
- ⚡ Event cameras react much faster to changes in the environment.